

Habib University
Digital Logic and Design
EE/CE-172/222
Assignment 3 Solution

Question: 01

Circuit Implementation.

2 pts

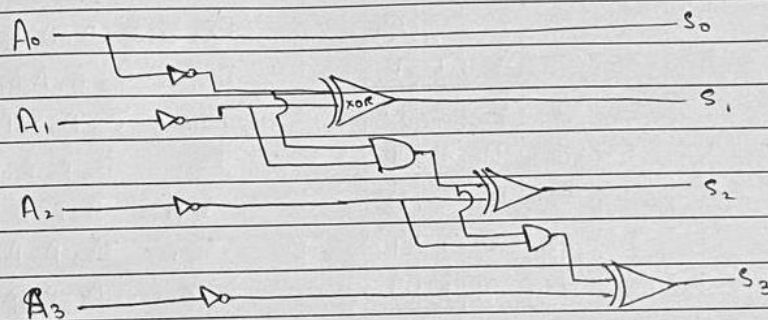
Let $A_3A_2A_1A_0$ be a 4-bit binary number and $S_3S_2S_1S_0$ be its 2's complement.

$$S_0 = A_0.$$

$$S_1 = A_1' \oplus A_0'$$

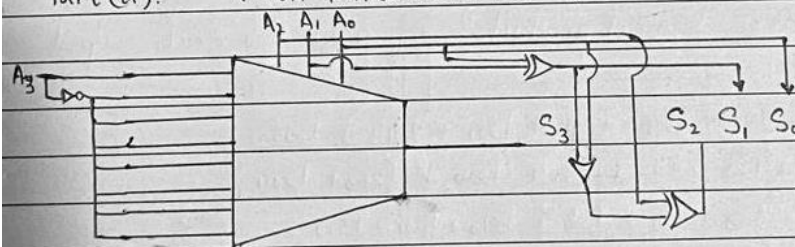
$$S_2 = A_2' \oplus (A_1'A_0')$$

$$S_3 = A_3' \oplus (A_2'A_1'A_0').$$



Part(a): (MUX IMPLEMENTATION)

6 pts



Date: _____

Part (b): (DECODER IMPLEMENTATION)

4 pts.

→ Reference Table for both part (a) & (b)

A_3	A_2	A_1	A_0	S_3	S_2	S_1	S_0	Let,
0	0	0	0	0	0	0	0	$D_0 = A_3' A_2' A_1' A_0'$
0	0	0	1	1	1	1	1	$D_1 = A_3 A_2 A_1 A_0$
0	0	1	0	1	1	1	0	$D_2 = A_3 A_2 A_1 A_0'$
0	0	1	1	1	1	0	1	$D_3 = A_3 A_2 A_1 A_0$
0	1	0	0	1	1	0	0	$D_4 = A_3 A_2 A_1 A_0'$
0	1	0	1	1	0	1	1	$D_5 = A_3 A_2 A_1 A_0$
0	1	1	0	1	0	1	0	$D_6 = A_3 A_2 A_1 A_0'$
0	1	1	1	1	0	0	1	$D_7 = A_3 A_2 A_1 A_0$
1	0	0	0	1	0	0	0	$D_8 = A_3 A_2 A_1 A_0'$
1	0	0	1	0	1	1	1	$D_9 = A_3 A_2 A_1 A_0$
1	0	1	0	0	1	1	0	$D_{10} = A_3 A_2 A_1 A_0'$
1	0	1	1	0	1	0	1	$D_{11} = A_3 A_2 A_1 A_0$
1	1	0	0	0	1	0	0	$D_{12} = A_3 A_2 A_1 A_0'$
1	1	0	1	0	0	1	1	$D_{13} = A_3 A_2 A_1 A_0$
1	1	1	0	0	0	1	0	$D_{14} = A_3 A_2 A_1 A_0'$
1	1	1	1	0	0	0	1	$D_{15} = A_3 A_2 A_1 A_0$

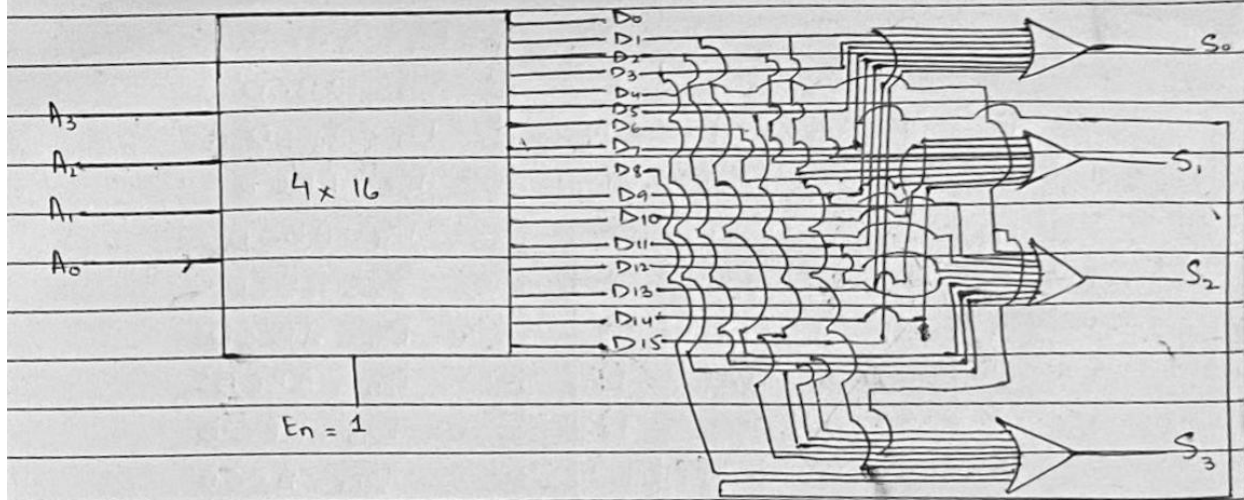
$$S_0 = D_1 + D_3 + D_5 + D_7 + D_9 + D_{11} + D_{13} + D_{15}$$

$$S_1 = D_1 + D_2 + D_5 + D_6 + D_9 + D_{10} + D_{13} + D_{14}$$

$$S_2 = D_1 + D_2 + D_3 + D_4 + D_9 + D_{10} + D_{11} + D_{12}$$

$$S_3 = D_1 + D_2 + D_3 + D_4 + D_5 + D_6 + D_7 + D_8$$

Date: _____



MUX implementation is better because it uses less gates and is less robust. 2 pts.

Question: 02

Part (a):

$$X = (AB' + A'B)C + A'BD'$$

2 pts

$$\Rightarrow X = ABC + A'BC + A'BD'$$

$$Y = AB + BD'$$

A	B	C	D	X	Y	
0	0	0	0	0	0	$D_0 = A'B'C'D'$
0	0	0	1	0	0	$D_1 = A'B'C'D$
0	0	1	0	0	0	$D_2 = A'B'CD'$
0	0	1	1	0	0	$D_3 = A'B'CD$
0	1	0	0	1	1	$D_4 = A'BC'D'$

8 pts

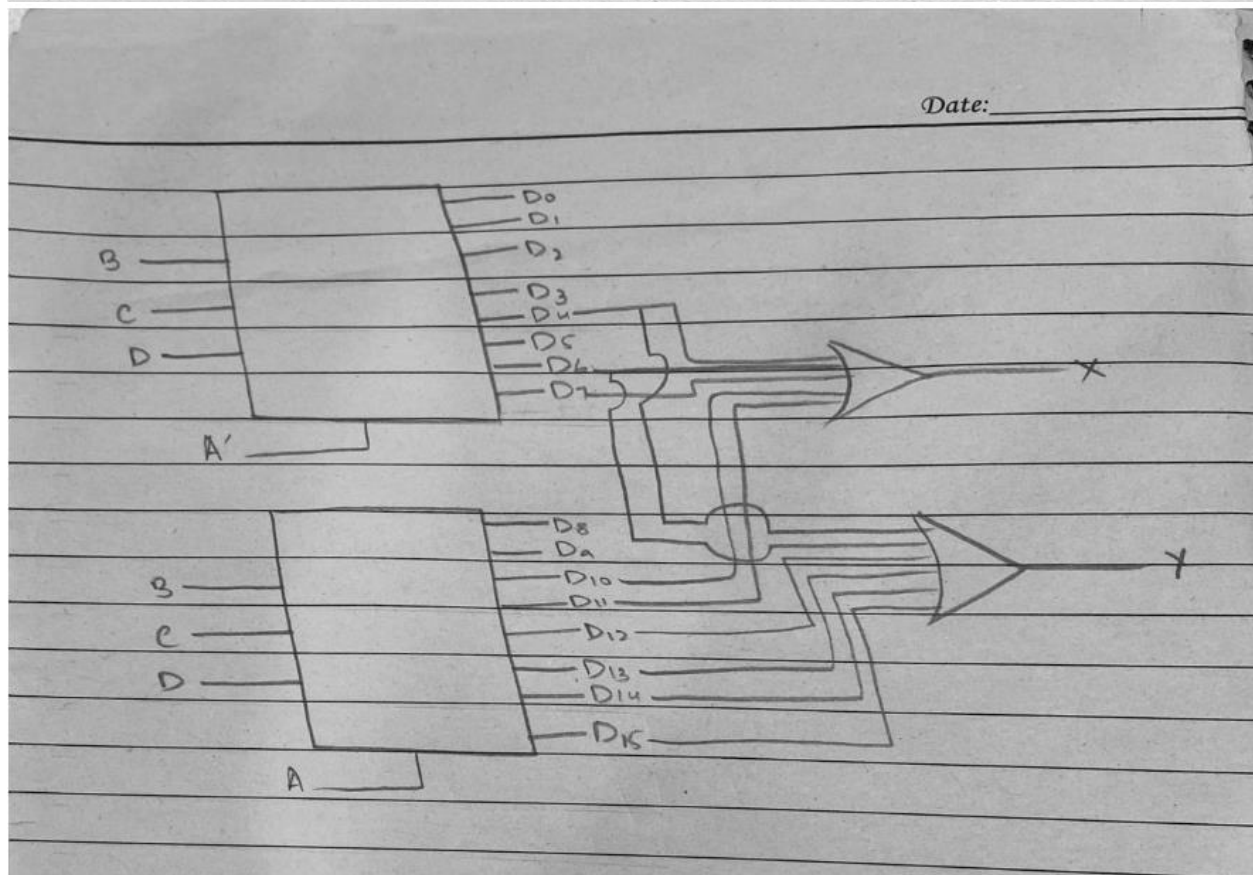
A	1	0	1	0	0	$D_5 = A'BC'D$
0	1	1	0	1	1	$D_6 = A'BCD'$
0	1	1	1	1	0	$D_7 = A'BCD$
1	0	0	0	0	0	$D_8 = ABC'D'$
1	0	0	1	0	0	$D_9 = ABC'D$
1	0	1	0	1	0	$D_{10} = ABC'D'$
1	0	1	1	1	0	$D_{11} = ABC'D$
1	1	0	0	0	1	$D_{12} = ABC'D'$
1	1	0	1	0	1	$D_{13} = ABC'D$
1	1	1	0	0	1	$D_{14} = ABCD'$
1	1	1	1	0	1	$D_{15} = ABCD$

Part (b):

6 pts.

$$X = D_4 + D_6 + D_7 + D_{10} + D_{11}$$

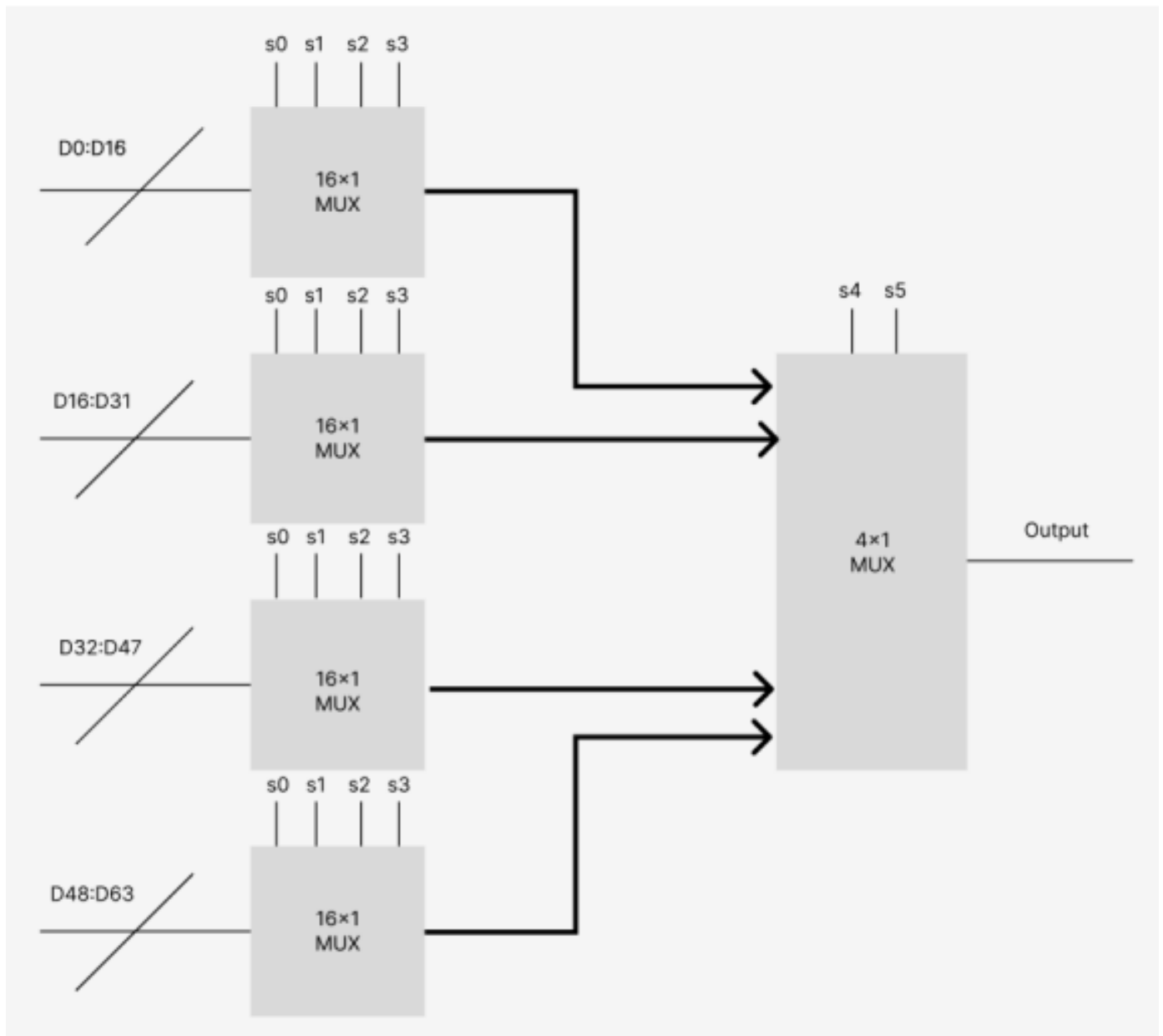
$$Y = D_4 + D_6 + D_{12} + D_{13} + D_{14} + D_{15}$$



Question: 03

Part (a):

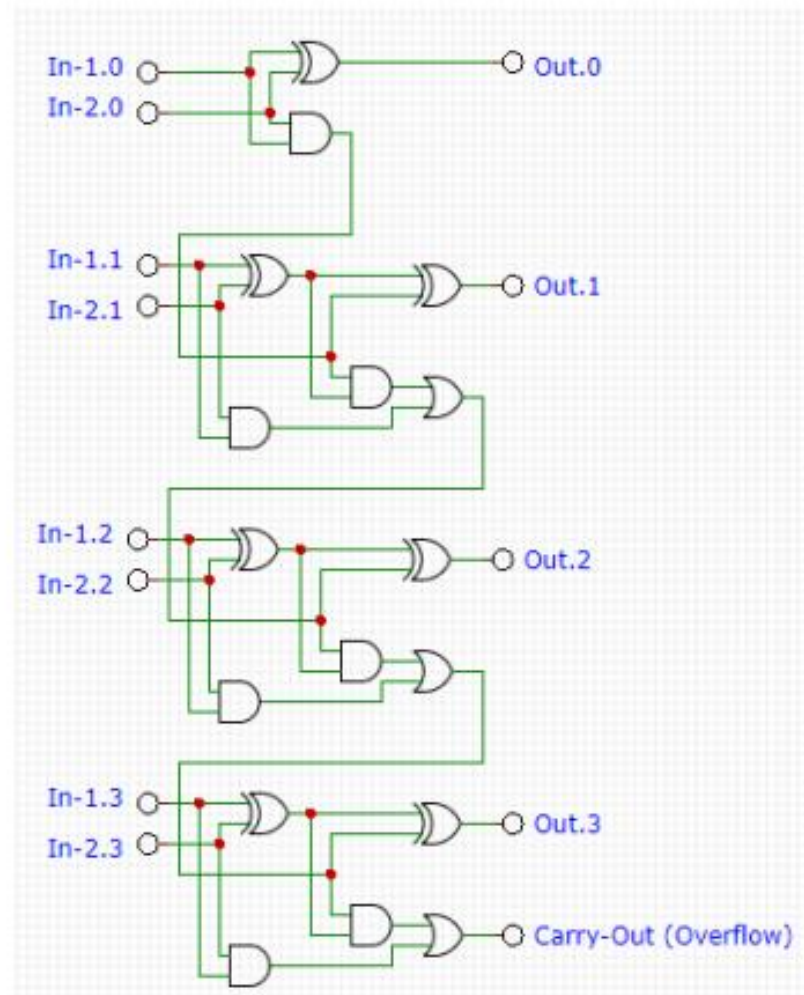
5 pts



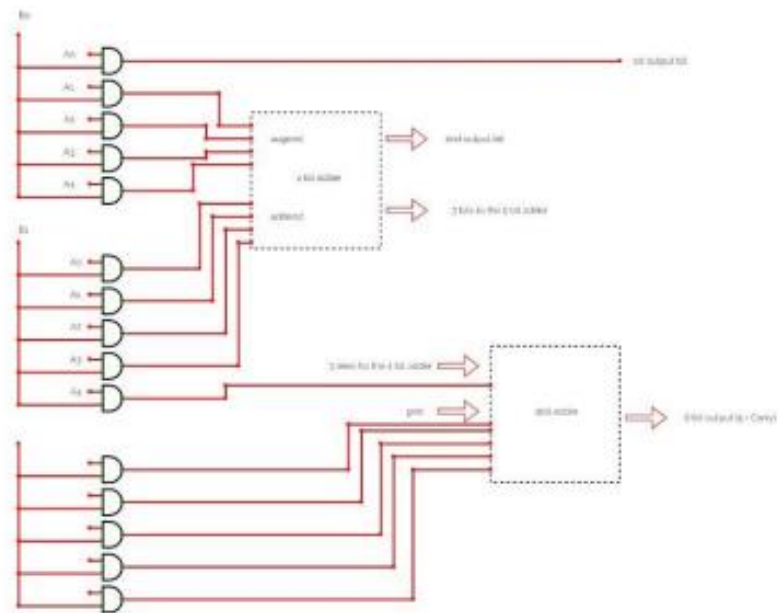
Part (b):

5 pts

b) Final binary expression must be obtained first before progressing towards the design.



Extend the circuit given above for one more bit.



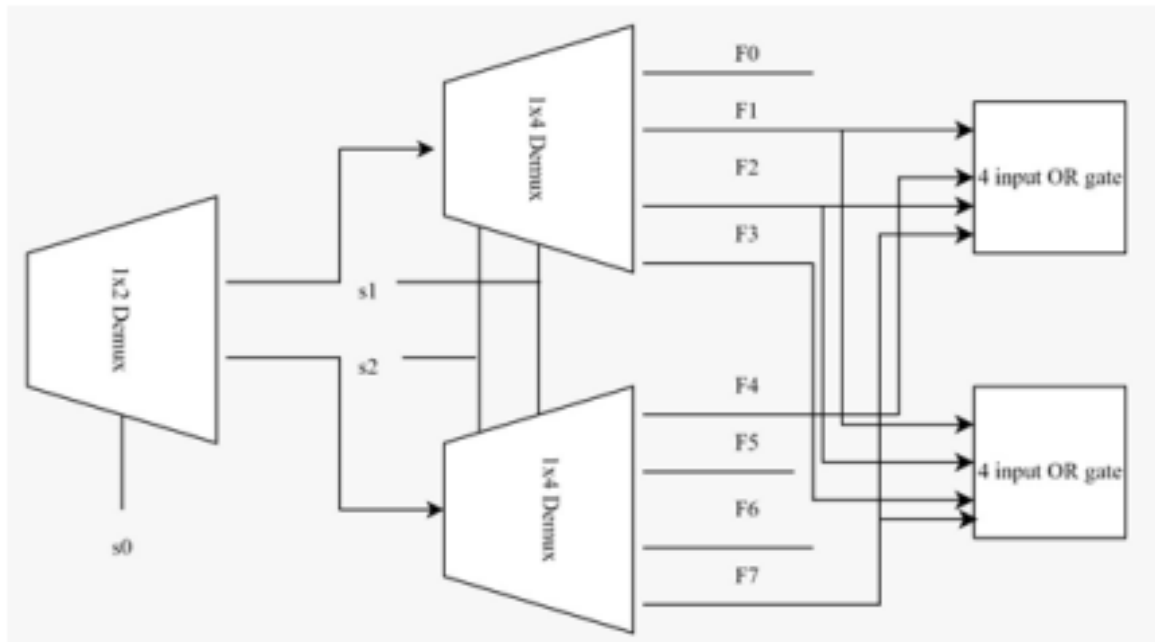
Part (c):

5 pts

c)

A	B	Cin	Y	Cout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

A	B	Cin	F0	F1	F2	F3	F4	F5	F6	F7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



Question: 04

10 pts

We find the 9's complement by subtracting each digit by 9. In this case, we use BCD digits and subtract them from 9. Getting W,X,Y,Z as follows:

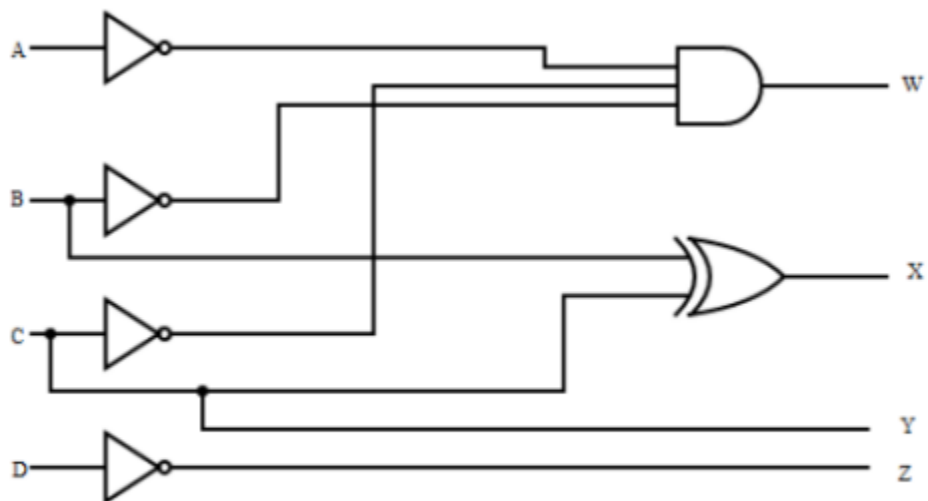
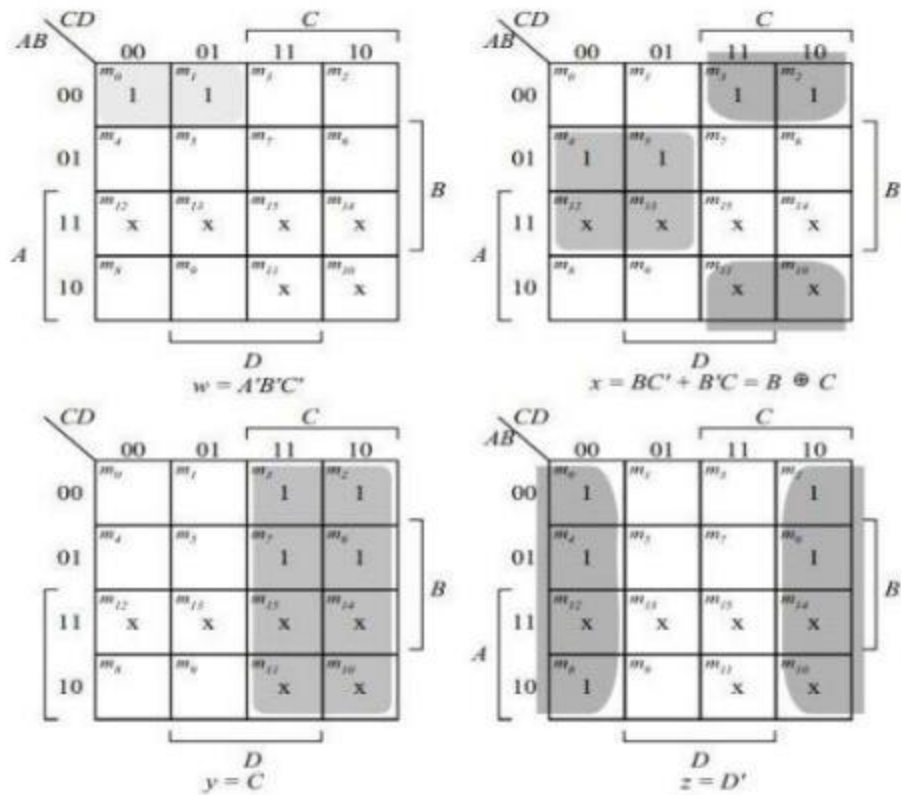
$$9-0 = 9$$

$$9-1 = 8$$

.....

We're working with BCD digits thus we don't care what comes in the table at 10 to 16 positions

	A	B	C	D	W	X	Y	Z
0	0	0	0	0	1	0	0	1
1	0	0	0	1	1	0	0	0
2	0	0	1	0	0	1	1	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	0	1
5	0	1	0	1	0	1	0	0
6	0	1	1	0	0	0	1	1
7	0	1	1	1	0	0	1	0
8	1	0	0	0	0	0	0	1
9	1	0	0	1	0	0	0	0
10	1	0	1	0	X	X	X	X
11	1	0	1	1	X	X	X	X
12	1	1	0	0	X	X	X	X
13	1	1	0	1	X	X	X	X
14	1	1	1	0	X	X	X	X
15	1	1	1	1	X	X	X	X



Question: 05

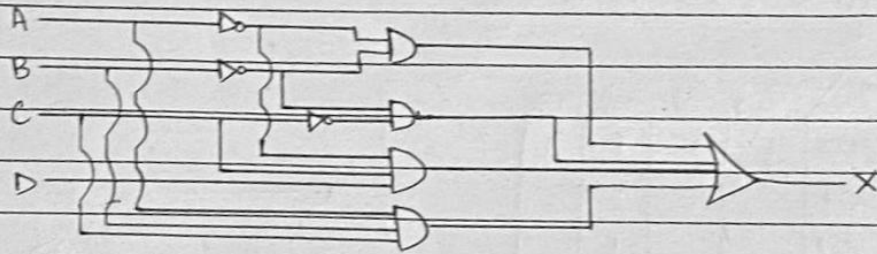
Part (a):

Weight Delta (gms)	A	B	C	D	X (motor brakes)	Y (aspirator function)	Z (dripper function)
+50	0	0	0	0	1	1	0
+37.5	0	0	0	1	1	1	0
+25	0	0	1	0	1	1	0
+31.25	0	0	1	1	1	1	0
-6.25	0	1	0	0	0	0	0
0	0	1	0	1	0	0	0
-12.5	0	1	1	0	0	0	0
-43.45	0	1	1	1	1	0	1
+22	1	0	0	0	1	1	0
+27.7	1	0	0	1	1	1	0
-32.5	1	0	1	0	0	0	0
-33	1	0	1	1	0	0	0
-25	1	1	0	0	0	0	0
-22	1	1	0	1	0	0	0
+43.45	1	1	1	0	1	1	0
+12.5	1	1	1	1	1	1	0

Part (b):**7 pts**

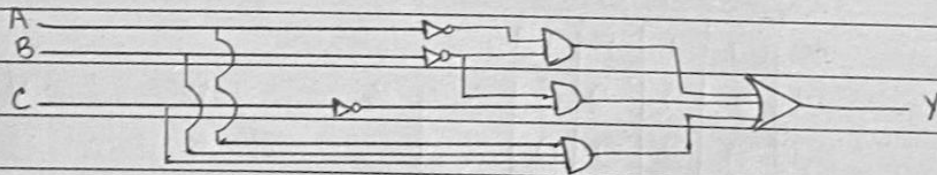
AB \ CD	00	01	11	10
00	1	1	1	1
01	0	0	1	0
11	0	0	1	1
10	1	1	0	0

$$X = A'B' + B'C' + A'CD + ABC$$

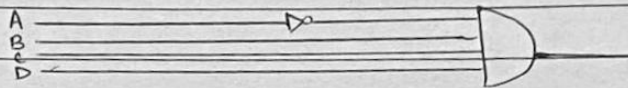


AB \ CD	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	1	1
10	1	1	0	0

$$Y = A'B' + B'C' + ABC$$



$$Z = A'B'CD$$



Part (c):

6 pts

Weight Delta (gms)	A	B	C	D	X (motor brakes)	Y (aspirator function)	Z (dripper function)
+50	0	0	0	0	1	1	0
+37.5	0	0	0	1	1	1	0
+25	0	0	1	0	1	1	0
+31.25	0	0	1	1	1	1	0
-6.25	0	1	0	0	0	0	0
0	0	1	0	1	0	0	0
-12.5	0	1	1	0	0	0	0
-43.45	0	1	1	1	1	0	1
+22	0	0	0	0	1	1	0
+27.7	0	0	0	1	1	1	0
-32.5	0	0	1	0	1	1	0
-33	0	0	1	1	1	1	0
-25	0	1	0	0	0	0	0
-22	0	1	0	1	0	0	0
+43.45	0	1	1	0	0	0	0
+12.5	0	1	1	1	1	0	1

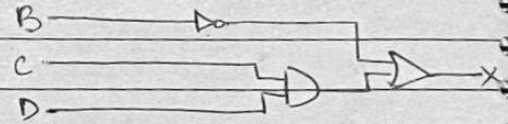
For understanding purpose, let's make k-map for each function given that $A=0$.

AB \ CD	00	01	11	10
00	1	1	1	1
01			1	
01			1	
00	1	1	1	1

$$X = A'B' + A'CD$$

Since A is always 0, $A' = 1$

$$\Rightarrow X = B' + CD$$

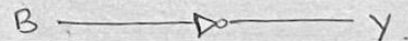


Similarly for Y,

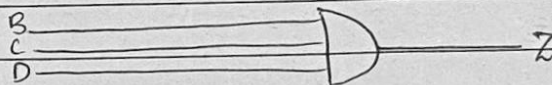
AB \ CD	00	01	11	10
00	1	1	1	1
01	0	0	0	0
01	0	0	0	0
00	1	1	1	1

$$Y = A'B'$$

$$\Rightarrow Y = B'$$



$$Z = BCD$$



Findings: When A is grounded, only BCD control the motor, the control logic becomes less accurate. More states map to the same actions.

Question: 06

Part (a):

5 pts

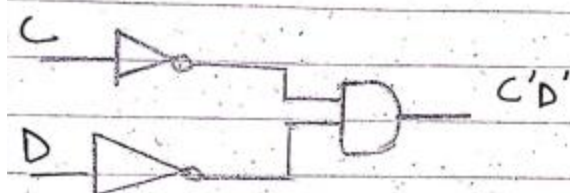
a)	A	B	C	D	Remainder	Y
	0	0	0	0	0	1
	0	0	0	1	1	X
	0	0	1	0	2	0
	0	0	1	1	3	0
	0	1	0	0	0	1
	0	1	0	1	1	X
	0	1	1	0	2	0
	0	1	1	1	3	0
	1	0	0	0	0	1
	1	0	0	1	1	X
	1	0	1	0	2	0
	1	0	1	1	3	0
	1	1	0	0	0	1
	1	1	0	1	1	X
	1	1	1	0	2	0
	1	1	1	1	3	0

Y = (0, 4, 8, 12)

Don't care conditions (1, 5, 9, 13)

CD \ AB	00	01	11	10
00	1	X		1
01	1	X		
11	1	X		
10	1	X		

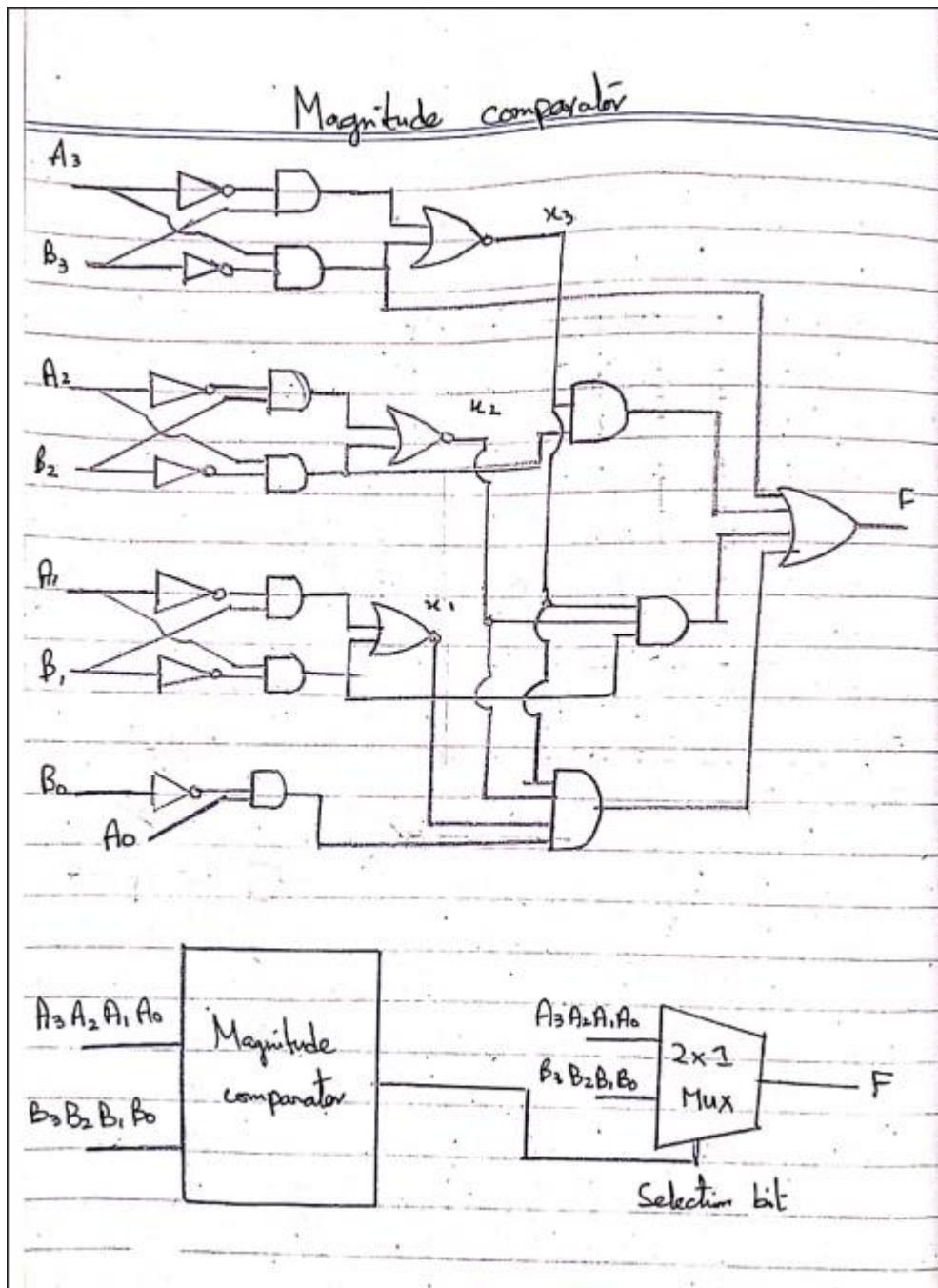
$$Y = C'D'$$



Part (b):

10 pts

A magnitude comparator will be used, which compares the two unique ID's and gives output 1 only the first ID is greater. For example 2 students have ID's $W_1X_1Y_1Z_1$ & $W_0X_0Y_0Z_0$ thus output is only 1 $W_1X_1Y_1Z_1$ is greater. The output will be used in MUX as selection bit.

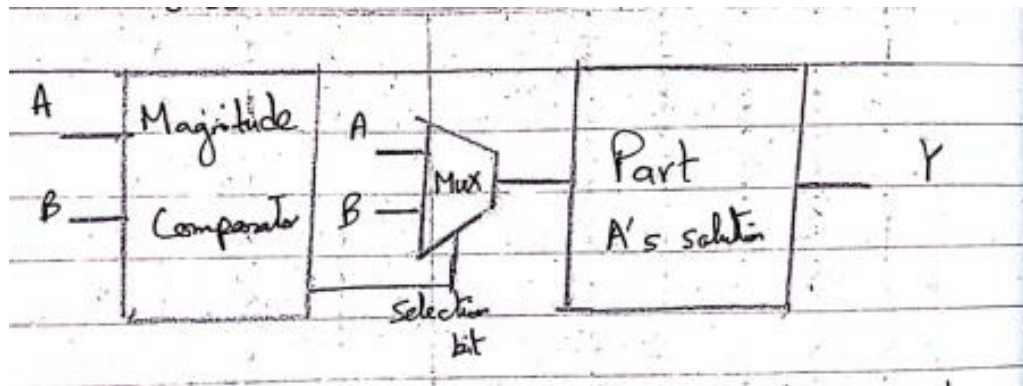


Magnitude comparator sketched above:-

$$F = A_3 B_3' + x_3 A_2 B_2' + x_3 x_2 A_1 B_1' + x_3 x_2 x_1 A_0 B_0'$$

Part (c):

5 pts



Part (d):

5 pts

If more than two students come at the same time, we can make the system bigger in two simple ways. One way is to use a **priority encoder**. We make a line for every possible ID number, then the encoder will automatically give us the highest ID that is present, which we then send to the Group-1 detector. The other way is to make a **comparator tree**. We compare students in pairs, pick the one with the higher ID, then compare the winners again until only one highest ID is left. That ID is then checked by the Group-1 detector. This way the same idea from parts (a) and (b) still works even if many students come at once.